Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Linear Independence
- Understanding Linear Independence in Vector Spaces — Reading • 18 min
- Linear Independence, Basis, and Rank — Ungraded App Item • 20 min
- Linear Independence — Reading • 30 min

Linear Mapping
- Exploring Transformations: Understanding Linear Mappings — Reading • 1h
- Linear Mapping: Part 1 — Video • 8 min
- Linear Mapping: Part 2 — Video • 8 min
- Linear Mapping — Reading • 5 min
- Matrix Representation — Reading • 6 min
- Practice Quiz: Linear Mapping — Graded Assignment • Grade: 100%

Lesson 3: Eigenvalues and Eigenvectors
- Introduction to Eigenvalues and Eigenvectors — Reading • 40 min
- Eigenvectors and Eigenvalues: Examples and Applications — Reading • 30 min

Lesson 4: Cholesky Decomposition
- Cholesky Decomposition — Reading • 5 min
- Solved Examples — Reading • 10 min

Lesson 5: Singular Value Decomposition
- Introduction to Singular Value Decomposition (SVD) — Reading • 28 min
- Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors — Reading • 15 min

Module 3
Probability Foundations for Statistical Learning

Lesson 1: Sum Rule, Product Rule, and Bayes' Theorem
- **Sum Rule, Product Rule, and Bayes' Theorem** — Reading • 15 min
- Sum Rule — Reading • 4 min
- Product Rule (Chain Rule) — Reading • 2 min
- Bayes' Theorem — Reading • 43 min

Lesson 2: Gaussian Distribution
- Univariate Gaussian Distribution — Reading • 4 min

Sum Rule, Product Rule, and Bayes' Theorem

0:04 / 1:08 2x

Read Section 6.3: "Sum Rule, Product Rule and Bayes' Theorem" from *Mathematics for Machine Learning* by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong.

In this section, you will explore the fundamental rules of probability that allow for calculating complex probabilities from simpler ones. The **sum rule** and **product rule** are essential for understanding joint and marginal probabilities, while **Bayes' theorem** is key to updating probabilities based on new information. Bayes' theorem is particularly valuable in machine learning for tasks involving inference, such as probabilistic classification and Bayesian networks.

Reflect on these questions while you read:

- How do the sum and product rules simplify complex probability calculations?
- Why is Bayes' theorem a fundamental tool for updating probabilities in light of new evidence?
- How are these rules applied in probabilistic machine learning models?
- In what machine learning applications can Bayes' theorem be particularly impactful?

Engaging with these questions will help you grasp the importance of these rules and their applications in machine learning.

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